

Nadhir Halbouni

Full-Stack Software Engineer - AI/LLM Applications

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🏠 Ariana - Tunisia

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Summary

Full-stack engineer specializing in **data-heavy SaaS systems**, **backend reliability**, and **automation**. Experienced in building and scaling production systems across multiple repositories, handling real-time features, payments, and growth mechanisms. Strong focus on **ownership**, **system design**, and **delivering end-to-end solutions** from database to user experience. Building **LLM-powered** apps with **agents** and **RAG**.

Experience

Drivago

Ariana, Tunisia

Software Engineer

September 2024 - Present

- Designed and built a **cross-repository training platform** spanning 7+ services, enabling secure multi-device usage and supporting concurrent usage without session conflicts.
- Integrated **payment gateways (Konnect, Flouci)**, implementing full payment lifecycle and automating subscription billing across thousands of transactions.
- Owned **production incident resolution** for 500+ client agencies, reducing incident resolution time by rapidly reproducing and debugging cross-platform issues.
- Managed **production deployment** on **Linux servers** using **Nginx and PM2**, maintaining uptime and overseeing environment configuration across all services.
- Engineered **automation systems (cron jobs)** for **scrapping**, **database backups**, and **recurring billing**, eliminating manual operations and saving hours of weekly maintenance work.
- Developed **authentication and security systems**, including **magic-link login**, OAuth (Google/Facebook), and **real-time fraud detection** enforcing single-session access.
- Led migration of **4 core repositories** from Angular v14 → v19, standardizing architecture and enabling full RTL/LTR support.
- Built a **referral & coupon system** driving user acquisition and retention through modular discount logic and reward tracking.

Drivago

Ariana, Tunisia

Software Engineer Intern

July 2024 - August 2024

- Built **BI dashboards** tracking SaaS metrics (MRR, ARR, churn, ARPU) to support product decisions.
- Designed and optimized **PostgreSQL queries** for retention analysis and cohort tracking.
- Developed advanced **segmentation & filtering systems** for backoffice operations (subscriptions, payments, activity).

Projects

AI Flow — LLM Chat Platform with Agents, RAG & Voice Input

August 2026 - Present

Full-stack AI assistant pairing a LangGraph **tool-calling agent** with document-grounded answers and push-to-talk voice input.

- Engineered a **LangGraph agent** with 7 tools (web search, calculator, weather, RAG over uploads, long-term memory), streaming **SSE responses** with live tool-call progress.
- Built a **RAG pipeline** (PDF/DOCX ingestion, chunking, pgvector similarity search) plus persistent conversation memory backed by a Postgres checkpointer.
- Added push-to-talk **voice input** via a **whisper.cpp** STT endpoint (ffmpeg normalization, timeout-guarded subprocesses, transcribe-to-draft UX).
- Tech: Python, FastAPI, LangGraph, React, PostgreSQL + pgvector, Ollama, whisper.cpp

Live Demo: [aiflow.n3dhir.com](#)

GitHub: [github.com/n3dhir/AI-Flow](#)

Conversational money tracker: log spending by messaging a Telegram bot, ask for summaries in your AI assistant, or check the web UI — one shared backend.

- Built a **Telegram bot** that logs expenses from everyday messages and **voice notes**, with **human-in-the-loop** confirmation before anything is saved.
- Connected the same backend to AI assistants via **MCP** (Claude, opencode) and to a web UI, so totals and spending history stay consistent everywhere.
- Designed private per-user accounts with revocable access tokens.
- Tech: TypeScript, Node.js, Express, MCP SDK, PostgreSQL, JWT, whisper.cpp, Telegram Bot API

Live Demo: buddy.n3dhir.com
GitHub: github.com/n3dhir/Buddy
Telegram Bot: t.me/buddy_n3dhir_bot

Web-based technical assessment platform featuring code execution, automated evaluation, and backend processing for candidate submissions.

- Built a full-stack **technical assessment platform** supporting MCQ, coding challenges, and open-ended evaluations.
- Integrated a **C++ code execution api** to run user-submitted code, with test case validation, automated scoring, and pass/fail evaluation.
- Designed backend evaluation pipeline handling **submissions, validation, and result aggregation**.
- Tech: Angular, Node.js, Express.js, MongoDB, ACE Editor

Live Demo: testify.n3dhir.com
Demo Video: drive.google.com/file/d/11PjLQudEmKk0j9YbJ3aseVqYhDcJsMyp

Skills

- **Core Strengths** Backend Systems, API Design, REST APIs, Authentication & Security, Payments Integration, Automation, Data Modeling
- **Backend & Data** Node.js, Express.js, TypeScript, Python, FastAPI, PostgreSQL, pgvector, Knex.js, MongoDB
- **Frontend** Angular (v14–v19), React, RxJS, Angular Material, SCSS, Tailwind CSS
- **Infrastructure & Tools** Linux, Nginx, PM2, Cron Jobs, ngrok, Git/GitHub
- **AI / LLM** LangGraph, LangChain, RAG, LLM Integration, Ollama, whisper.cpp, MCP
- **Other** SSR, SSG, SEO Optimization, OAuth (Google/Facebook)

Education

Software Engineering Degree 2023–2026
Higher Institute of Informatics (ISI), Ariana, Tunisia

Computer Science Degree 2020–2023
Higher Institute of Informatics (ISI), Ariana, Tunisia

Certifications

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| • SQL (Advanced) Certificate — HackerRank | June 2023 |
| • Problem Solving (Intermediate) Certificate — HackerRank | August 2022 |
| • Relational Database — freeCodeCamp | June 2023 |
| • JavaScript Algorithms and Data Structures — freeCodeCamp | June 2023 |
| • Algorithmic Toolbox — Coursera | May 2022 |
| • Meta Hacker Cup 2022 | September 2022 |

Languages

Arabic (Native), English (Professional), French (Professional), German (Basic)